

Last name _____

First name _____

LARSON—MATH 310—CLASSROOM WORKSHEET 15
Matrix-Matrix Multiplication.

Review

1. Let A be the 3×2 matrix: $\begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix}$ We showed that $f(\vec{x}) = A\vec{x}$ defines a *linear* function $f: \mathbb{R}^2 \rightarrow \mathbb{R}^3$. We first check that f is *additive* then that f is *homogeneous*.

Systems of Linear Equations (Chp. 8)

1. Rewrite the following linear system in matrix notation.

$$\begin{array}{rclcl} x & + & 2y & + & 3z & = & 6 \\ 2x & + & 3y & + & 4z & = & 9 \\ 3x & + & 4y & + & 5z & = & 10 \end{array}$$

Matrix Multiplication (Chp. 10)

Let

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \\ 3 & 1 & 2 \end{bmatrix}$$
$$B = \begin{bmatrix} 1 & 2 \\ 2 & 3 \\ 3 & 1 \end{bmatrix}$$

2. Why is AB well-defined?
3. Why is BA *not* well-defined?
4. What will the size of AB be?
5. Find AB using the *definition* of matrix multiplication.

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \\ 3 & 1 & 2 \end{bmatrix}$$

$$B = \begin{bmatrix} 1 & 2 \\ 2 & 3 \\ 3 & 1 \end{bmatrix}$$

6. Find AB using the *column interpretation* of matrix multiplication.

7. Find AB using the *row interpretation* of matrix multiplication.

8. Find $(AB)^T$.

9. Find A^T .

10. Find B^T .

11. Find $B^T A^T$. Check that $(AB)^T = B^T A^T$.